

Gradient-Sensitive Functional Descriptors for Interferometric Surface Metrology: Complementing Amplitude-Averaging ISO Parameters

Abstract

Amplitude-averaging surface parameters such as R_a may show limited response to localised defects — a controlled sensitivity experiment indicates that a single 100 nm synthetic spike injected into a spherical-artefact profile produces no detectable change in R_a at the reconstruction resolution, and the ISO 4287 shape parameters R_{sk} and R_{ku} show similarly limited response to such symmetric localised features. This paper evaluates the same reconstructed profile data within a functional-analytic norm framework — L^p , Sobolev $W^{1,2}$, and bounded-variation (BV) norms, standard tools of functional analysis here applied within a metrological processing pipeline — finding higher sensitivity to such defects (BV density: +18%; $W^{1,2}$ gradient seminorm: +622%) without additional hardware. A complete proof-of-concept pipeline from raw Michelson-interferometer fringe images to nanometre profile descriptors is presented and validated on two JENOPTIK standards (roundness standard FN 111, $\varnothing = 29.9588$ mm, 24,000 frames; magnification standard FN 101 with Flick flat, 100,000 frames), each measured in five 400° sweeps. External validation against 104 tactile reference measurements (Hommel Roundscan 535) yields $E_n = 0.61$ for the FN 111 (optical–tactile polar comparison: Supplementary Fig. S12) and 0.3% agreement for the FN 101, suggesting metrological consistency of the optical pipeline within the stated within-run type-A uncertainties. The functional-norm quantities are presented as complementary to the ISO 4287 baseline: R_{sk} responds to asymmetry, BV density responds to localised gradients, and their combination may provide descriptive sensitivity beyond that of any single clas-

sical parameter in the tested scenarios. The central finding is that BV density detects a certified geometric feature (the Flick flat on the FN 101 standard) that remains below the detection threshold of amplitude-averaging parameters in a sliding-window analysis matched to the feature’s certified angular extent, illustrating — in the tested proof-of-concept scenarios — that gradient-based quantities can provide complementary descriptive sensitivity to localised geometric features without additional hardware; the L^4/L^1 peakiness ratio (1.39 ± 0.08 for the FN 111, 1.53 ± 0.03 for the FN 101; difference not statistically distinguishable at 95% confidence given overlapping expanded uncertainties) is reported as an illustrative distribution-shape indicator reflecting the amplification of dominant profile excursions relative to R_a .

Keywords: interferometry, roundness, surface profile, functional norms, bounded variation, Sobolev seminorm, uncertainty

1. Introduction

Surface topography is one of the most important properties of technical parts. The accuracy of form and roughness influences contact conditions, wear and lifetime. Roundness, understood as deviation from a perfect circle, remains a central requirement in precision manufacturing; Zhu et al. emphasised that precise measurements are needed both for form tolerances and for tribological surface texture assessment [1]. At nanometric scale, such measurements require high-precision optical systems [2]. The foundational reference for surface metrology terminology and parameter definitions is the handbook of Whitehouse [3].

Optical dimensional metrology has a long connection with interference [4]. Modern optical surface metrology relies on diffraction, polarisation and interference [5, 6]. Diffraction and scattering have been applied to roughness measurement [7–11], while interference remains central for ultra-precise surface-form measurement [12, 13] and phase-shifting interferometry is widely used in topography evaluation [14–16].

Several interferometric instruments have been proposed for high-accuracy

17 surface and roundness measurement. Polack et al. described a Michelson-based
 18 system for surface-shape determination built around a high-resolution stitching
 19 interferometer developed by Thomasset et al. [17, 18]. Fan et al. presented
 20 a roundness measuring system for microspheres based on two small Michel-
 21 son interferometers [19]. Kuhnel et al. proposed an interferometric system for
 22 roundness and cylindricity measurement of ring gauges with 0.1 nm resolution
 23 over a large measurement range [20]. Commercial instruments can be highly ac-
 24 curate, but they are often costly and may not expose the full acquisition chain
 25 needed for experimental method development.

26 The present work uses a self-built, non-contact Michelson-type roundness
 27 measurement system assembled from off-the-shelf components. The optical
 28 setup, apparatus description and radial image-processing route for phase extrac-
 29 tion were published in a previous study [21], where only roughness parameters
 30 were reported for the ceramic standard FN 111. The present work extends that
 31 study in three directions: (i) roundness (RON_t , RON_q) is evaluated for the first
 32 time with this self-built optical setup, (ii) a second artefact — the JENOPTIK
 33 magnification standard FN 101 with a Flick flat — is measured and validated
 34 against its DAkkS-DKD certificate, and (iii) a controlled sensitivity-comparison
 35 methodology is introduced: synthetic defects of known geometry are injected
 36 into measured profiles, and the response of classical ISO parameters is com-
 37 pared against that of L^p , Sobolev $W^{1,2}$, and bounded-variation (BV) norms
 38 — standard tools of functional analysis here applied to interferometric profile
 39 metrology — all computed from the same nanometre residual. The comparison
 40 framework, rather than any individual norm, constitutes the methodological
 41 contribution.

42 The computational problem is that modern interferometric surface metrol-
 43 ogy records long image sequences before a numerical profile exists. In the present
 44 data, however, each PNG file is already a separate interferogram in which the
 45 phase is encoded spatially by the bright and dark fringes inside the image. The
 46 files in the FN 111 and FN 101 folders are therefore not treated as a temporal
 47 phase-shifting stack of one fixed scene. Instead, each image $I_i(x, y)$ is processed

independently to obtain one wrapped excess-fraction value ϵ_i . The ordered sequence of these values is then unwrapped over image order and converted to a nanometre profile. The full measurement archive contains 24,000 grayscale interferograms of the FN 111, with image size 648×488 px, and 100,000 grayscale interferograms of the JENOPTIK magnification standard FN 101, with image size 640×480 px.

The literature relevant to this comparison is mature but only partly connected. Interferometric testing has developed approaches for stitching, registration, motion-error compensation, and systematic-error control [22–26]. Repeatability is commonly evaluated through RMS/PV quantities or pixel-wise standard-deviation matrices [27, 28]. Uncertainty-oriented surface metrology embeds surface-texture parameters in GUM-style propagation [29]. Functional data analysis provides a statistical language for function-valued observations [30, 31], and scale-dependent roughness studies show that characterisation can depend on bandwidth and filtering scale [32–34].

The research gap is therefore specific. Existing interferometric literature provides robust instrumentation and phase reconstruction; standards-oriented work provides scalar profile and form parameters with uncertainty evaluation; and functional data analysis provides a mathematical language for function-valued observations. What has received less attention is a controlled, within-profile comparison of the sensitivity of classical ISO amplitude parameters against variation-based functional norms, computed from the same nanometre residual and validated against an external tactile reference. The present work addresses this by: (i) injecting synthetic defects of known geometry into measured profiles to quantify descriptor response under controlled conditions; (ii) applying the full pipeline to two DAkkS-DKD-certified artefacts at large scale (24,000 and 100,000 interferograms) with a self-built radial excess-fraction interferometer; and (iii) benchmarking the optical reconstruction against 104 tactile reference measurements. Areal parameters are not reported because the single-image radial extraction does not produce a calibrated areal height map. The aim is to assess whether L^p , bounded-variation, and Sobolev seminorms, evaluated

79 alongside ISO 4287 parameters on the same reconstructed residual profile, offer
80 complementary descriptive sensitivity to controlled profile perturbations, and
81 to evaluate the within-run repeatability of all quantities across five recorded
82 sweeps.

83 2. Experimental Details

84 2.1. Optical Setup

85 The optical layout of the measurement system is shown in Fig. 1. The
86 system is a two-channel modified Michelson interferometer with channels shifted
87 by approximately 90° in phase. When an array detector is used instead of a
88 point detector, the second channel is not strictly required [35]. In this system it
89 was retained because it is useful for object centring and can serve as a control
90 channel.

91 The optical setup and the full apparatus-level description were reported in a
92 previous study on radial image processing for phase extraction in rough-surface
93 interferometry [21]. They are repeated here in condensed form to make the
94 present functional-descriptor analysis self-contained and to document the ac-
95 quisition conditions behind the FN 111 and FN 101 interferogram archives.

96 The system was first tested with the JENOPTIK roundness standard FN 111
97 (photographs: Supplementary Fig. S1) (Al_2O_3 , $\varnothing = 29.9588$ mm) [36]. A single-
98 mode 4.5 mW diode laser with wavelength 532 nm was used as the light source.
99 A current-stabilised supply powered the laser to obtain a stable wavelength [37].
100 The laser beam passes through a polariser (P in Fig. 1) to set the polarisation
101 of the light at 45° to the setup plane, minimising the difference in reflectivity
102 between the two channels.

103 Key components: achromatic lenses $L_{1,2,4,5}$, non-polarising beamsplitter
104 BS , $\lambda/8$ liquid phase retarder, $\lambda/10$ reference mirror M , DIN-4 microscope
105 objective L_3 (NA 0.10), polarising beamsplitter PBS , and two Gigabit Ether-
106 net CCD cameras (640×480 px, 120 fps).

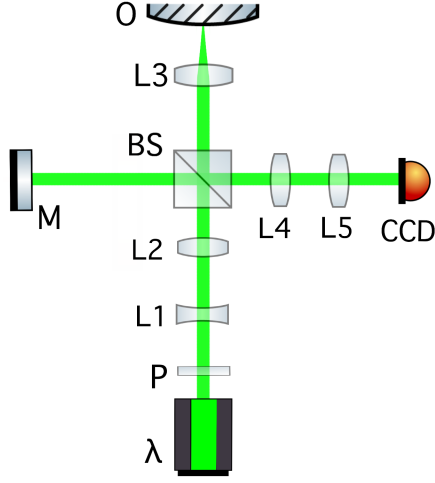


Figure 1: Optical layout of the two-channel modified Michelson interferometer: λ – diode laser (532 nm); P – linear polariser; L – lenses; L_3 – microscope objective; M – reference mirror ($\lambda/10$ flatness); BS – non-polarising beamsplitter; O – object on rotation stage; CCD – camera.

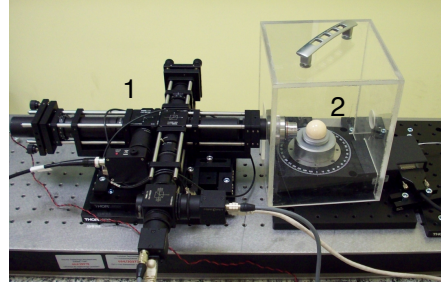


Figure 2: The complete measurement system: (1) modified Michelson interferometer in cage mounting, (2) high-precision rotation stage with the artefact.

107 The interferometer was built in a cage system for thermal, acoustic and me-
 108 chanical stability. A plexiglass box shields the measurement object, and the
 109 structure is mounted on an area translation stage. A high-precision rotation
 110 stage (encoder accuracy 2.0 mdeg, typical wobble $\pm 12 \mu\text{rad}$, typical eccentric-
 111 ity $\pm 0.40 \mu\text{m}$; full specifications: Supplementary Table S1) rotates the object.
 112 Both interferometer and stage are mounted on a honeycomb breadboard with
 113 vibration isolation.

114 Measurements were performed at night (23:00–05:00) to minimise building-
 115 activity disturbances. The setup was thermally and acoustically shielded (plex-
 116 iglass box, wool blanket) and placed on a granite plate on layered wooden-
 117 polystyrene vibration isolation (photographs: Supplementary Fig. S2). Five
 118 semiconductor temperature sensors ($\pm 0.2^\circ\text{C}$) monitored stability; the tempera-
 119 ture near the reference and object arms remained stable to $\pm 0.2^\circ\text{C}$ over six-hour

120 acquisitions.

121 2.2. Control System and Acquisition Workflow

122 Cameras were controlled by FLIR SDK-based C programs and Bash scripts [38].
 123 Capture time was recorded per frame for angular-position verification. The ro-
 124 tation stage was controlled via USB ASCII commands [39]; a 25 s settling delay
 125 ensured constant velocity before acquisition began. Object centring was per-
 126 formed manually using the two-camera phase-shifted configuration at 45° rota-
 127 tion increments. Fringe evaluation used modified open-source R code [35, 40].
 128 Table 1 lists the acquisition parameters for both archives; five 400° sweeps were
 129 recorded per artefact, with the 40° overlap retained for profile-end comparison.

Table 1: Measurement and sampling parameters of the interferometric acquisition workflow.

Parameter	FN 111	FN 101
Rotation speed, ω	$0.50^\circ/\text{s}$	$0.08^\circ/\text{s}$
CCD frame rate	6 fps	4 fps
Recorded sweeps, n	5	5
Total frames, N	24,000	100,000
Angular step, $\Delta\alpha = \omega/\text{fps}$	$0.083^\circ/\text{frame}$	$0.020^\circ/\text{frame}$
Frames per degree	12	50
Frames per useful 360°	4320	18,000
Frames per recorded 400° sweep	4800	20,000
Overlap per sweep	$40^\circ / 480 \text{ frames}$	$40^\circ / 2000 \text{ frames}$
Total acquisition time	4000 s	25,000 s

130 The object was measured in one continuous five-sweep rotation without repo-
 131 sitioning. Each recorded sweep spans 400° ; successive useful 360° segments are
 132 shifted by 40° , giving a natural between-sweep angular offset. The 40° overlap
 133 region provides a model-free repeatability check: RMS height difference between
 134 consecutive sweeps is 12 nm (FN 111) and $0.18 \mu\text{m}$ (FN 101), consistent with
 135 Table 2.

136 3. Image Processing and Functional-Descriptor Method

137 3.1. Processing Pipeline

138 The pipeline follows the radial excess-fraction method of a previous study [21]
 139 with added sinusoidal refinement. Each interferogram $I_i(x, y)$ is decoded inde-
 140 pendently to a wrapped excess fraction ϵ_i ; the sequence is unwrapped chronolog-
 141 ically, converted to nanometres ($z_i = \nu_i \lambda / 2$), and reduced by the object-specific
 142 LSCI reference model. Sweep boundaries are retained for repeatability analysis.

143 3.2. Step-by-Step Radial Phase Decoding

144 For each interferogram $I_i(x, y)$, a radial intensity profile about the fringe
 145 centre is extracted and B-spline-smoothed; bright-ring maxima are detected in
 146 D^2 space, where circular-fringe theory gives $D_{ij}^2 = c_i(j + \epsilon_i - 1)$ for the j th ring
 147 of diameter D_{ij} . A linear fit $D_{ij}^2 = \alpha_i j + \beta_i$ yields the wrapped excess fraction
 148 $\epsilon_i = (1 + \beta_i / \alpha_i) \bmod 1$ (2). Where the radial fringe signal permits ($R^2 \geq 0.65$,
 149 phase within 0.20 fringe of the ring-diameter estimate), a sinusoidal refinement
 150 is applied; otherwise the ring-diameter estimate is retained. The ordered series
 151 ϵ_i is unwrapped chronologically to a continuous fringe-order coordinate ν_i , and
 152 the nanometre profile follows from $z_i = \nu_i \lambda / 2$ with $\lambda = 532$ nm (one fringe
 153 order $\equiv 266$ nm) (3).

$$154 \quad D_{ij}^2 = c_i(j + \epsilon_i - 1), \quad (1)$$

$$155 \quad \epsilon_i = \left(1 + \frac{\beta_i}{\alpha_i}\right) \bmod 1. \quad (2)$$

$$156 \quad z_i = \nu_i \frac{\lambda}{2}, \quad \lambda = 532 \text{ nm}. \quad (3)$$

156 3.3. Topography Profile, Reference Removal and Eccentricity Reduction

157 After phase-to-height conversion, the roundness profile is extracted following
 158 ISO 12181-1 [41] (which supersedes the withdrawn ISO 4291). For each 360°
 159 segment, a least-squares reference circle (LSCI) is fitted to the angular height
 160 profile $z_i = h(\theta_i)$ in Cartesian coordinates:

$$(x_i, y_i) = ((R_0 + z_i) \cos \theta_i, (R_0 + z_i) \sin \theta_i), \quad (4)$$

161 where $R_0 = 14.9794$ mm is the nominal radius of the JENOPTIK FN 111. The
 162 LSCI centre (x_c, y_c) minimises $\sum_i [\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} - R_0]^2$, yielding the
 163 centred radial residuals ρ_i . The LSCI fit is numerically equivalent to removing
 164 the first harmonic of the angular profile, but it provides the formal ISO 4291
 165 reference circle required for roundness metrology. After LSCI centring, a phase-
 166 correct Gaussian low-pass profile filter with 50% transmission at 15 undulations
 167 per revolution (UPR) is applied to the residual profile, consistent with the filter
 168 specification used for the tactile reference measurements and with the Gaussian
 169 profile filter defined in ISO 16610-21. All classical and gradient-based descriptors
 170 are computed from this LSCI-centred, Gaussian-filtered residual $\rho_i^{(\text{filt})}$.

171 For the JENOPTIK magnification standard FN 101, the same processing
 172 chain is applied: a linear trend is removed from the unwrapped phase se-
 173 quence before height conversion (Sec. 5.1), the LSCI reference circle is fitted
 174 to the angular profile, and the Gaussian low-pass filter (50% at 15 UPR) is
 175 applied. Although both artefacts are processed through identical filtering steps,
 176 the 15 UPR cutoff corresponds to different spatial wavelengths on the circum-
 177 ference: $2\pi \times 14.9794$ mm/15 = 6.27 mm for the FN 111 ($\varnothing = 29.9588$ mm) and
 178 $2\pi \times 15$ mm/15 = 6.28 mm for the FN 101 ($\varnothing \approx 30$ mm) — effectively identical.
 179 The angular sampling differs substantially (0.083° vs. 0.020° , Table 1), but the
 180 Gaussian filter sigma is set by the UPR cutoff, not by the sampling interval, so
 181 the filtered profiles are directly comparable.

182 3.4. Classical Profile and Roundness Metrics

183 All classical quantities are evaluated on nanometre residuals, not on raw
 184 intensities, following the profile method of ISO 4287 [42]. Areal parameters
 185 (ISO 25178 [43]) are not reported because the single-image radial extraction
 186 does not produce a calibrated areal height map. For a residual profile or vector
 187 of residual pixels p , the reported profile-style roughness descriptors are

$$R_a = \frac{1}{M} \sum_{j=1}^M |p_j - \bar{p}|, \quad R_q = \left[\frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^2 \right]^{1/2}, \quad (5)$$

188 and

$$R_z = \max_j (p_j - \bar{p}) - \min_j (p_j - \bar{p}). \quad (6)$$

189 The profile skewness R_{sk} and kurtosis R_{ku} — ISO 4287 shape parameters that
 190 characterise asymmetry and peakedness of the height distribution — are also
 191 reported:

$$R_{sk} = \frac{1}{R_q^3} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^3, \quad R_{ku} = \frac{1}{R_q^4} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^4. \quad (7)$$

192 $R_{sk} = 0$ for a symmetric distribution; $R_{sk} < 0$ indicates a predominance of
 193 valleys over peaks. $R_{ku} = 3$ for a normal distribution; $R_{ku} > 3$ indicates
 194 a spikier (leptokurtic) profile. These are the only ISO 4287 parameters that
 195 capture distribution shape beyond amplitude averages, and their inclusion allows
 196 a direct test of whether gradient-based descriptors provide information beyond
 197 even the best ISO shape parameters. For the eccentricity-corrected ceramic
 198 angular profile $q_C(\theta)$,

$$RONt = \max_{\theta} q_C(\theta) - \min_{\theta} q_C(\theta), \quad RONq = \left[\frac{1}{2\pi} \int_0^{2\pi} q_C^2(\theta) d\theta \right]^{1/2}. \quad (8)$$

199 These are profile/form quantities in nanometres. B-spline pre-smoothing (20 de-
 200 grees of freedom, $\approx 18^\circ$ period) stabilises the LSCI fit; the subsequent Gaussian
 201 low-pass filter (50% at 15 UPR, ISO 16610-21) defines the metrological band-
 202 width. All descriptors are computed from the same Gaussian-filtered residual.

203 3.5. Functional Gradient-Based Descriptors

204 The concept of a complete normed vector space originates from the founda-
 205 tional work of Banach [44]. For discrete profile data of finite length N , the vector
 206 space $\mathbb{R}^N$ equipped with any norm is a Banach space; the present work does
 207 not invoke infinite-dimensional completeness but draws on the taxonomy of L^p ,
 208 Sobolev, and bounded-variation norms that Banach’s framework systematised.
 209 For a centred residual profile $p(s)$ and $1 \leq p < \infty$,

$$\|p\|_p = \left(\frac{1}{L} \int_0^L |p(s)|^p ds \right)^{1/p}, \quad \|p\|_\infty = \max_{s \in [0, L]} |p(s)|. \quad (9)$$

210 The reported functional descriptors include L^1 , L^2 , L^∞ , the one-dimensional
 211 Sobolev $W^{1,2}$ seminorm and the bounded-variation (BV) total variation,

$$|p|_{W^{1,2}} = \left(\int_0^L \left| \frac{dp}{ds} \right|^2 ds \right)^{1/2}, \quad TV(p) = \int_0^L \left| \frac{dp}{ds} \right| ds. \quad (10)$$

212 In discrete form, for a uniformly sampled profile p_j ($j = 1, \dots, N$) with angular
 213 spacing $\Delta\alpha$,

$$|p|_{W^{1,2}} \approx \left(\Delta\alpha \sum_{j=1}^{N-1} \left(\frac{p_{j+1} - p_j}{\Delta\alpha} \right)^2 \right)^{1/2}, \quad TV(p) \approx \sum_{j=1}^{N-1} |p_{j+1} - p_j|. \quad (11)$$

214 Both $W^{1,2}$ and BV density carry sampling-interval dependence and require
 215 conversion to angular units ($\text{nm deg}^{-1/2}$ and nm deg^{-1} , respectively) for cross-
 216 instrument comparison; “nm sample $^{-1}$ ” is retained for the fixed-acquisition setup.

217 The curvature RMS (root-mean-square of the discrete second derivative;
 218 reported in Supplementary Table S2) carries a $(\Delta\alpha)^{-2}$ sampling dependence and
 219 should not be compared across instruments with different angular resolution.

220 3.6. Repeatability Uncertainty

221 Each dataset contains five 400° sweeps recorded in a single continuous ro-
 222 tation without stopping or repositioning. The five useful 360° segments are ex-
 223 tracted from this continuous sequence; because each recorded sweep spans 400° ,
 224 successive useful revolutions are shifted by 40° on the object. These segments
 225 are not strictly independent — they share the same continuous-run conditions
 226 (laser warm-up state, stage-bearing temperature, ambient drift). The standard
 227 deviation across sweeps therefore reflects within-run repeatability rather than
 228 between-run reproducibility. For a descriptor X evaluated on each sweep, the
 229 mean value is $\bar{X}$ and the within-run standard uncertainty is estimated as

$$u_{\text{within}}(X) = \frac{s_X}{\sqrt{n}}, \quad n = 5, \quad (12)$$

230 where s_X is the sample standard deviation across the five sweeps. The reported
 231 expanded uncertainty is

$$U_{95}(X) = t_{0.975, n-1} u_{\text{within}}(X). \quad (13)$$

232 With $n = 5$, $t_{0.975,4} = 2.776$; the χ^2_4 -based 95% CI for the standard deviation
233 spans $[0.6s_X, 2.9s_X]$, so U_{95} is indicative rather than definitive. Because the five
234 segments share continuous-run conditions (laser state, stage temperature, ambi-
235 ent drift), the $1/\sqrt{n}$ factor in u_{within} may underestimate the true uncertainty of
236 the mean; all reported uncertainties are *lower bounds*. A full GUM-compliant
237 budget (type-B terms: laser wavelength stability, refractive-index variations,
238 Abbe error [45], camera nonlinearity, form-removal uncertainty) is beyond the
239 present scope. With $n = 5$ sweeps and the observed within-run variability, the
240 minimum detectable relative difference between two descriptors (80% power,
241 $\alpha = 0.05$, two-sided paired t -test) is approximately 30% for the FN 111 and
242 5% for the FN 101, consistent with the exploratory character of the descrip-
243 tor comparison. A per-sweep trend of FN 111 $RONt$ values gives a slope of
244 $8.4 \pm 4.1 \text{ nm sweep}^{-1}$ ($p = 0.12$), consistent with zero drift, though the sweep-
245 to-sweep scatter cautions that U_{95} may underestimate long-term reproducibility.

246 3.7. Computational Implementation

247 The reconstruction workflow (`scripts/reconstruct_radial_profile_sequence.`
248 `py`) decodes ϵ_i from each interferogram, unwraps the sequence, converts to
249 nanometres, applies LSCI fitting and Gaussian low-pass filtering (50% at 15 UPR,
250 ISO 16610-21), and exports CSV/JSON metrics with sweep-repeatability uncer-
251 tainty. Vector PDF figures document each processing stage. The pipeline is in-
252 voked as `PYTHONPATH=src.venv/bin/pythonscripts/reconstruct_radial_profile_`
253 `sequence.py` with dataset-specific parameters (full command in the data repos-
254 itory). Per-image excess-fraction decoding and unwrapping require approxi-
255 mately 0.5 ms per frame (measured on an Intel Core i7-9700K, single-threaded
256 Python 3.9), so the full 124,000-frame archive processes in under two minutes
257 on consumer hardware.

258 4. Results

259 The results follow the updated radial sequence route only. Each raw im-
260 age is first decoded to a wrapped excess fraction from its radial fringe pat-

tern. The ordered excess-fraction series is then unwrapped in chronological order, converted to nanometres, and corrected for eccentricity. Finally, classical profile/roundness descriptors and functional gradient-based descriptors are computed from the same nanometre residual profiles.

4.1. Radial Phase Decoding and Nanometre Profile Reconstruction

The FN 111 folder contained 24,000 interferograms and the steel folder contained 100,000 interferograms. The reported run used full-resolution images, the central radial aperture of 240 px, B-spline-smoothed peak detection, continuity-aware EFM peak-subset tracking and an EFM prominence threshold of 0.04 of the radial fringe-signal range. All five recorded sweeps were processed.

Figs. S3-S6 (single-image EFM and sequence processing steps) are provided in the Supplementary Materials.

4.2. Roughness, Roundness and Functional Descriptors

Table 2 gives the numerical comparison after LSCI reference circle and Gaussian low-pass filtering (50% at 15 UPR). Both artefacts are processed through the same filtering chain. Each descriptor is reported as the full-sequence value together with the 95% expanded type-A repeatability uncertainty estimated from the five sweeps. Per-sweep values for all descriptors are provided in Supplementary Table S3.

The numerical identity $RONt = R_z$ for both artefacts (Table 2) reflects the fact that after LSCI centring and Gaussian low-pass filtering (50% at 15 UPR), the residual profile is dominated by a single lobe (harmonic decomposition: Supplementary Fig. S19); the peak-to-valley roundness deviation and the profile roughness peak-to-valley therefore coincide. This identity is specific to the present artefact-filter combination and should not be expected to hold for artefacts with multi-lobed form deviations or for different filter cutoffs.

4.3. Classical-Functional-Descriptor Comparison

The numerical identity $R_a = L^1$ and $R_q = L^2$ (Table 2) confirms that both descriptor families are evaluated on the same nanometre residual. For the

Table 2: Full-resolution radial EFM reconstruction from raw interferograms (LSCI + Gaussian low-pass, 50% at 15 UPR, ISO 16610-21). Uncertainties are 95% expanded within-run repeatability estimates from five sweeps ($n = 5$, $t_{0.975,4} = 2.776$). Units: nm for all descriptors except BV density and $W^{1,2}$ gradient (nm sample⁻¹). Certified *RONt*: FN 111 244 ± 52 nm (Hommel Roundscan 535, $n = 104$); FN 101 $41.94 \mu\text{m}$ (form tester, $n = 12$). The FN 101 certificate does not report an expanded uncertainty; therefore only the percentage difference is reported for this artefact rather than an E_n score.

Descriptor	FN 111	FN 101
R_a	$66.3 \pm 4.7 \text{ nm}$	$8.525 \pm 0.115 \mu\text{m}$
R_q	$77.6 \pm 5.0 \text{ nm}$	$10.376 \pm 0.090 \mu\text{m}$
R_z	$278 \pm 22 \text{ nm}$	$42.07 \pm 0.48 \mu\text{m}$
<i>RONt</i>	$278 \pm 22 \text{ nm}$	$42.07 \pm 0.48 \mu\text{m}$
<i>RONq</i>	$77.6 \pm 5.0 \text{ nm}$	$10.376 \pm 0.090 \mu\text{m}$
L^1 mean	$66.3 \pm 4.7 \text{ nm}$	$8.525 \pm 0.115 \mu\text{m}$
L^2 RMS	$77.6 \pm 5.0 \text{ nm}$	$10.376 \pm 0.090 \mu\text{m}$
L^∞	$150 \pm 12 \text{ nm}$	$21.97 \pm 0.42 \mu\text{m}$
BV total	$1.137 \pm 0.062 \mu\text{m}$	$118.4 \pm 6.2 \mu\text{m}$
BV density	$0.263 \pm 0.014 \text{ nm sample}^{-1}$	$6.58 \pm 0.35 \text{ nm sample}^{-1}$
$W^{1,2}$ grad.	$0.310 \pm 0.015 \text{ nm sample}^{-1}$	$8.76 \pm 0.43 \text{ nm sample}^{-1}$
<i>ISO shape parameters</i>		
R_{sk} (skewness)	-0.09 ± 0.04	0.00 ± 0.09
R_{ku} (kurtosis)	1.98 ± 0.21	2.51 ± 0.08
<i>Derived norm ratios</i>		
L^4/L^1 peakiness	1.39 ± 0.08	1.53 ± 0.03
<i>External validation</i>		
E_n score	0.61	< 1 (0.3% diff.; cf. Fig. S12)

290 FN 111, BV total variation carries 1.2% relative expanded uncertainty versus
 291 5.5% for R_a and 8.5% for R_z , suggesting that gradient-based descriptors may ex-
 292 hibit metrological stability comparable to or better than peak-referenced rough-
 293 ness parameters in the tested datasets. Comparative bar charts and uncertainty
 294 analyses are provided in the Supplementary Materials (Figs. S7–S8). Ceramic
 295 and steel profile reconstruction plots are shown in Supplementary Figs. S9–S10.

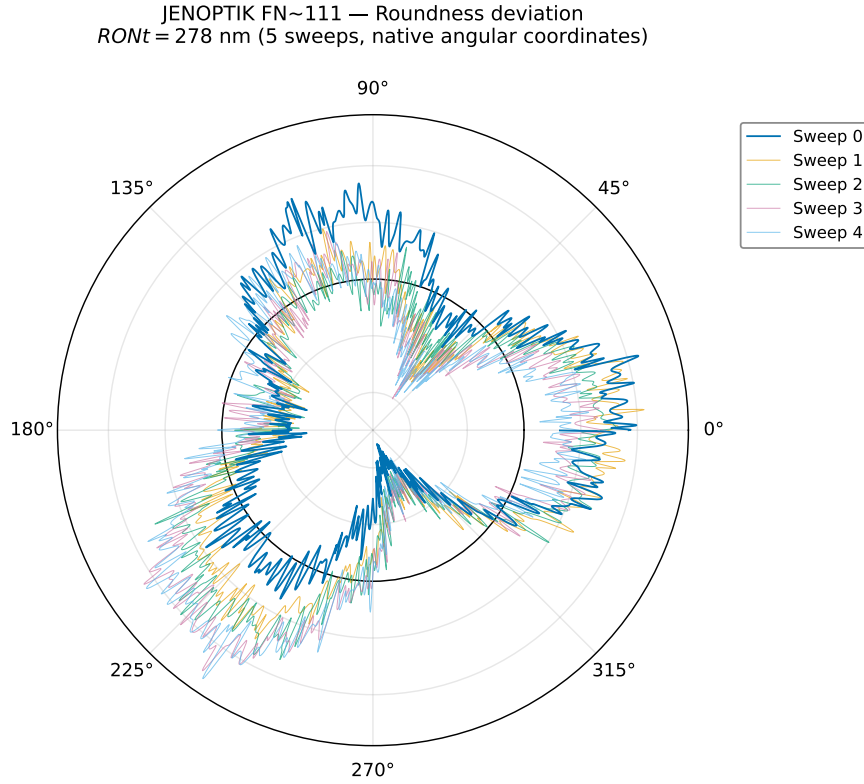


Figure 3: Ceramic roundness in polar coordinates (LSCI + Gaussian low-pass, 50% at 15 UPR). Five sweeps at native positions. $RONt = 278 \pm 22$ nm.

296 The steel roundness polar representation (FN 101, Flick flat) is provided in

the Supplementary Materials (Fig. S11).

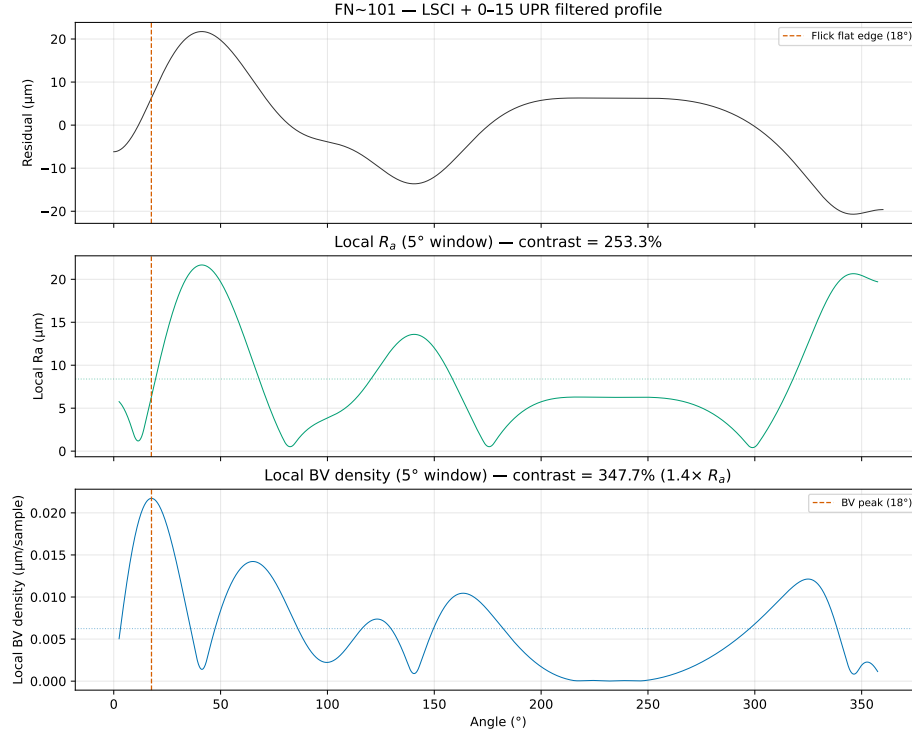


Figure 4: Flick flat detection on the FN 101 magnification standard. BV density (bottom) shows a clear spike at the Flick flat angular position, whereas R_a computed in a sliding window (top) shows no detectable response at the reconstruction resolution. The sliding-window width was matched to the certified angular extent of the Flick flat as specified in the DAkKS-DKD certificate (Art. 532 528); identical window parameters were used for both descriptors.

The Flick flat produces a clear response in BV density while showing no detectable response in a sliding-window R_a analysis at the reconstruction resolution (Fig. 4), illustrating the potential complementary sensitivity of gradient-based descriptors to localised geometric features.

302 5. Discussion

303 5.1. Confirmed Results

304 The analysis rests on a key metrological distinction: an interferogram is
305 an intensity measurement, not a phase map and not a height map. The half-
306 wavelength relation $z = \nu\lambda/2$ becomes valid only after each image has been
307 decoded to a wrapped excess fraction and the ordered excess-fraction sequence
308 has been unwrapped to the continuous fringe-order coordinate ν . This is why
309 the reported comparison uses only reconstructed nanometre profiles, not raw
310 greyscale images and not areal height maps.

311 The most sensitive part of the workflow is the EFM localisation of bright-
312 ring maxima, with an excess-fraction uncertainty of approximately ± 0.05 fringe
313 orders (tightened to ± 0.03 by sinusoidal refinement in 72% of FN 111 frames).
314 A single ϵ_i error of 0.05 fringe corresponds to 13 nm, sub-dominant to the
315 85 nm residual-profile RMS. The chronological unwrapping criterion rejects
316 phase jumps exceeding 0.35 fringe. The median number of detected rings is
317 three for both artefacts within the 240 px central aperture. The five 400° ce-
318 ramic sweeps require a geometric correction: successive useful 360° intervals are
319 shifted by 40° because the physical object angle advances continuously; without
320 this correction, the LSCI fit appears inconsistent even when phase extraction
321 within each sweep is repeatable.

322 The linear drift of $0.0246 \text{ fringe frame}^{-1}$ in the steel unwrapped phase se-
323 quence (versus 0.00021 for the FN 111) is attributed to cumulative bias in the
324 global unwrapping algorithm rather than thermal effects or axial runout [45].
325 The lower fringe contrast of the FN 101 interferograms — likely due to surface
326 finish — reduces per-frame EFM quality; the pipeline applies automatic linear
327 detrending where needed.

328 5.2. Interpretation

329 5.2.1. Eccentricity removal and wobble verification

330 Roundness descriptors are computed after LSCI reference-circle fitting (Sec. 3.3),
331 which removes the first harmonic. A shared model across all five sweeps yields

332 $R^2 = 0.99991$ with per-sweep amplitude variation of 63 nm (0.18%); the manufacturer-
 333 specified wobble ($\pm 12 \mu\text{rad}$) bounds the observed variation an order of magni-
 334 tude below $RONt = 278$ nm. The polar representation (Fig. 3) shows five
 335 overlaid sweeps repeatable to ± 50 nm, illustrating that the residual is domi-
 336 nated by low-order form deviation rather than random roughness.

337 5.3. Comparison with ISO Parameters

338 A central question of this work is whether gradient-based functional descrip-
 339 tors provide additional descriptive sensitivity alongside classical ISO parameters
 340 when applied to the same reconstructed profiles. To answer this rigorously, every
 341 descriptor was computed on the same reconstructed nanometre residual profiles.

342 **Mathematical equivalences.** The L^1 norm is defined as $\frac{1}{N} \sum |p_i - \bar{p}|$,
 343 which is identical to the definition of R_a . The L^2 norm is $\sqrt{\frac{1}{N} \sum (p_i - \bar{p})^2}$,
 344 identical to R_q . Computing both on the same centred residual yields *exact*
 345 *numerical equality*: $R_a - L^1 = 0$ and $R_q - L^2 = 0$ to machine precision for
 346 both artefacts (Table 2). These equalities are not empirical findings; they are
 347 definitional consequences. Labelling the arithmetic mean of absolute deviations
 348 as L^1 rather than R_a provides no new metrological information and should not
 349 be presented as a methodological advance.

350 It is emphasised that neither the L^p norms nor the Sobolev and BV semi-
 351 norms are presented as novel mathematical constructs; they are standard tools
 352 whose definitions are restated for completeness. Their value in the present con-
 353 text is threefold. First, the functional-analytic taxonomy places ISO parameters
 354 at specific positions (L^1 , L^2 , L^∞) within a larger space of possible descriptors,
 355 revealing that descriptors outside the ISO set — BV density, $W^{1,2}$ seminorm
 356 — respond differently to the same perturbations. Second, the synthetic-defect
 357 injection methodology provides a controlled, like-for-like comparison framework
 358 that can be applied to any descriptor computed from the same residual profile;
 359 the framework itself, rather than any individual norm, constitutes the method-
 360 ological contribution. Third, the Flick flat on the certified FN 101 standard pro-
 361 vides a concrete existence proof: a real, DAkkS-DKD-certified artefact whose

defining geometric feature is detected unambiguously by BV density while remaining below the detection threshold of sliding-window R_a (Fig. 4).

Extrema: R_z versus L^∞ . $R_z/L^\infty \approx 2$ for both artefacts, consistent with symmetric positive and negative excursions. L^∞ captures the single largest one-sided excursion; the ratio serves as a simple symmetry indicator.

Variation-based descriptors. A controlled sensitivity analysis (Supplementary Table S2) examined five synthetic defect types injected into the FN 111 residual profile. Amplitude-averaging descriptors (R_a , R_q , R_z) showed no detectable change at the reconstruction resolution — a single 100 nm spike changed R_a by less than 0.05%, below the per-frame phase-extraction precision — while BV density increased by +18% and the $W^{1,2}$ gradient seminorm by +622% for the same spike (Fig. 5). ISO shape parameters responded to asymmetry (R_{sk} changed by −1104% for a step) but not to symmetric localised features. R_{sk} and BV density appear complementary in the tested scenarios: R_{sk} responds to asymmetry, BV density responds to localised gradients.

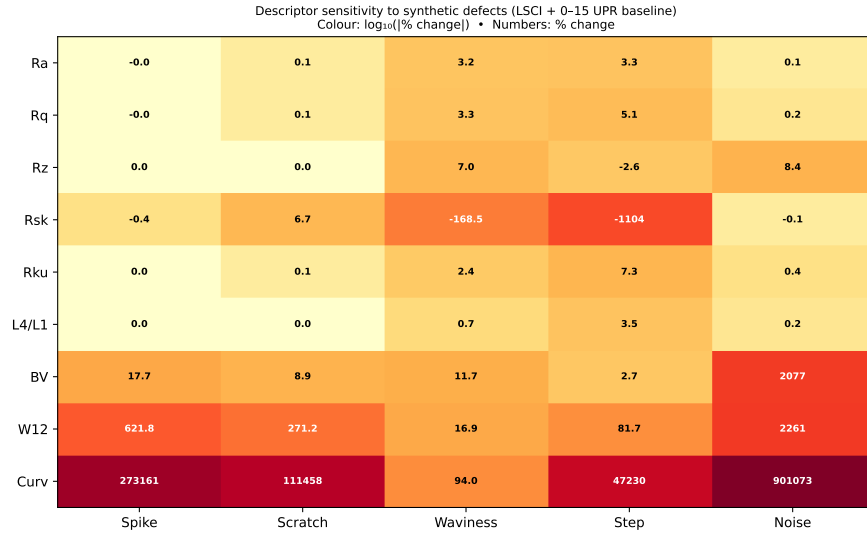


Figure 5: Sensitivity of ISO and gradient-based descriptors to five synthetic defect types injected into the FN 111 residual profile. Percentage change from the unperturbed baseline; bold labels indicate changes exceeding 10%.

Supporting analyses — descriptor correlation matrices, power spectral density comparison, Cohen’s d effect sizes, critical descriptor assessment, and L^p norm convergence — are provided in the Supplementary Materials (Figs. S13–S17). Given the number of descriptors compared (~ 15 across two artefacts), the reported differences should be interpreted as exploratory rather than confirmatory; no formal multiple-comparison correction has been applied.

Practical considerations. A practical suggestion for interferometric profile and roundness metrology is to retain R_a , R_q , R_z , R_{sk} , $RONt$ and $RONq$ as the primary engineering layer compliant with ISO 4287, and to consider supplementing them with BV density (in angular units, nm deg^{-1}) as a gradient-sensitive functional-analysis complement. The $W^{1,2}$ gradient seminorm is correlated with BV density in the present datasets (within the FN 111: Pearson $r = 0.95$, $n = 5$ sweeps; within the FN 101: $r = 0.93$, $n = 5$). The pooled $r = 0.94$ (95% CI $[0.72, 0.99]$, $n = 10$) is reported for descriptive purposes only — the sweeps are not independent observations and this pooled correlation should be interpreted with caution. $W^{1,2}$ may add limited independent information beyond BV density alone given the small sample. The L^4/L^1 peakiness ratio provides a supplementary distribution-shape axis with a direct physical interpretation as the amplification of dominant profile excursions relative to R_a . These additional descriptors are computed from the same residual profile without additional hardware or changes to the EFM pipeline. The cross-instrument validation suggests that gradient-based descriptors are applicable to any 1D profile, whether from interferometry or tactile roundness measurement.

Planned versus exploratory comparisons. The synthetic-defect responses (Supplementary Table S2, Fig. 5) and the Flick flat detection (Fig. 4) were planned comparisons motivated by the study’s objective of assessing gradient-based descriptor sensitivity. The correlation matrices, power spectral density comparison, Cohen’s d effect sizes, L^p norm convergence analysis, and the cross-descriptor stability comparison (Supplementary Figs. S13–S17) are exploratory and are intended to characterise the descriptor space rather than to test specific hypotheses. This distinction should be borne in mind when interpreting the

408 reported associations. The present study is a proof-of-concept demonstration;
 409 generalisability to other artefact types, materials, and measurement instruments
 410 remains to be established.

411 5.4. Limitations

412 Several limitations should be acknowledged. First, the uncertainty analysis
 413 is restricted to within-run Type A repeatability from five sweeps in a single
 414 continuous rotation. The small $n = 5$ limits statistical power (relative dif-
 415 ferences below $\sim 30\%$ cannot be resolved), and between-run reproducibility
 416 (independent setups, thermal cycles, re-centring) has not been assessed. A full
 417 GUM-compliant uncertainty budget, incorporating type-B evaluations of laser
 418 wavelength stability, refractive-index variations, Abbe error, camera nonlinear-
 419 ity, and form-removal uncertainty, is beyond the scope of this study; the reported
 420 expanded uncertainties should be interpreted as lower bounds.

421 Second, the validation is restricted to two artefacts from a single manufac-
 422 turer (JENOPTIK), both ~ 30 mm diameter; generalisability to other sizes,
 423 materials, or surface finishes has not been tested. Third, the synthetic-defect
 424 analysis uses idealised mathematical defects (isolated spike, uniform scratch,
 425 pure sinusoid) injected into the FN 111 ceramic profile only; the response to real
 426 manufacturing defects with complex morphologies may differ, and the defect-
 427 to-background ratio for the FN 101 is approximately 500 times smaller, so the
 428 detection sensitivity reported here may not generalise to artefacts with substan-
 429 tially larger form deviations. Fourth, BV density and $W^{1,2}$ seminorm carry
 430 intrinsic sampling-interval dependence requiring conversion to angular units
 431 for cross-instrument comparison. Fifth, the Flick flat result (Supplementary
 432 Fig. S18) is demonstrated on a single artefact with a known macroscopic feature;
 433 performance for subtler, micrometre-scale defects has not been characterised.

434 5.5. Metrological Implications

435 The results carry several implications for profile and roundness metrology
 436 practice. The finding that BV density responds to localised gradients that am-
 437 plitude averaging may not capture suggests — subject to further validation —

438 a potential role for gradient-based descriptors in applications where isolated
439 defects are of concern; possible examples include quality control of bearing sur-
440 faces, optical components, or precision seals. Because these descriptors are
441 computed from the same residual profile used for ISO 4287 parameters, their
442 adoption does not require changes to existing measurement acquisition proto-
443 cols.

444 The ISO 4287 framework remains the appropriate baseline for surface tex-
445 ture specification and conformity assessment. The gradient-based descriptors
446 discussed here are proposed as optional supplementary indicators, not as re-
447 placements for standardised parameters. Their metrological utility would be
448 strengthened by inter-laboratory comparisons, by evaluation on a wider range
449 of artefact types, and by the development of agreed procedures for converting
450 sampling-dependent units to instrument-independent angular gradient units.
451 Until such steps are taken, BV density and related descriptors should be re-
452 garded as research-grade tools whose diagnostic value has been observed under
453 controlled laboratory conditions on two specific artefacts.

454 5.6. Future Work

455 The present study suggests several directions for further investigation. (i) A
456 between-run reproducibility study with independent setups and re-centring would
457 establish whether the gradient-based descriptors retain their relative stability
458 under realistic laboratory conditions. (ii) Extending the validation to artefacts
459 with different diameters, materials (e.g., glass, carbide), and surface finishes
460 would test the generalisability of the findings. (iii) A systematic characterisation
461 of BV density response to real manufacturing defects (scratches, pits, inclusions)
462 of controlled dimensions would bridge the gap between the present synthetic-
463 defect analysis and industrial inspection scenarios. (iv) An inter-laboratory
464 comparison using a common artefact set and a standardised processing pipeline
465 would provide the reproducibility data needed to assess whether BV density
466 could eventually be considered for standardisation. (v) The development of
467 agreed angular-unit conventions for sampling-dependent gradient descriptors

would facilitate cross-instrument comparison and potential inclusion in reference software for roundness evaluation.

Data Availability

The analysis code, processed results, and all figures are available at <https://github.com/dawidkucharski/banach-profile-metrology> (release v1.1.0). The raw interferometric PNG archives (~32 GB) are available from the corresponding author upon reasonable request.

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